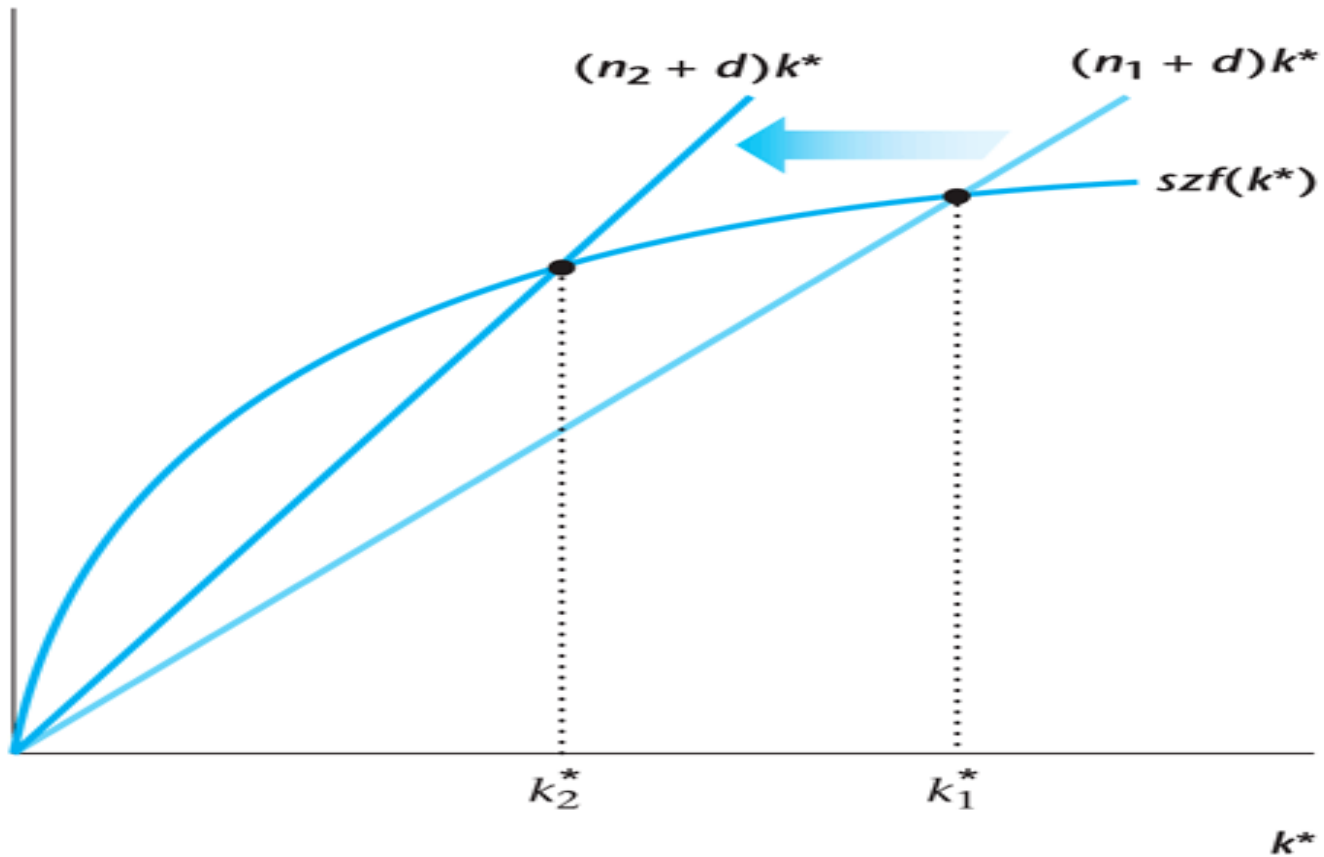


International Economic Development

ECON 4660/8666

Lecture 5. Population

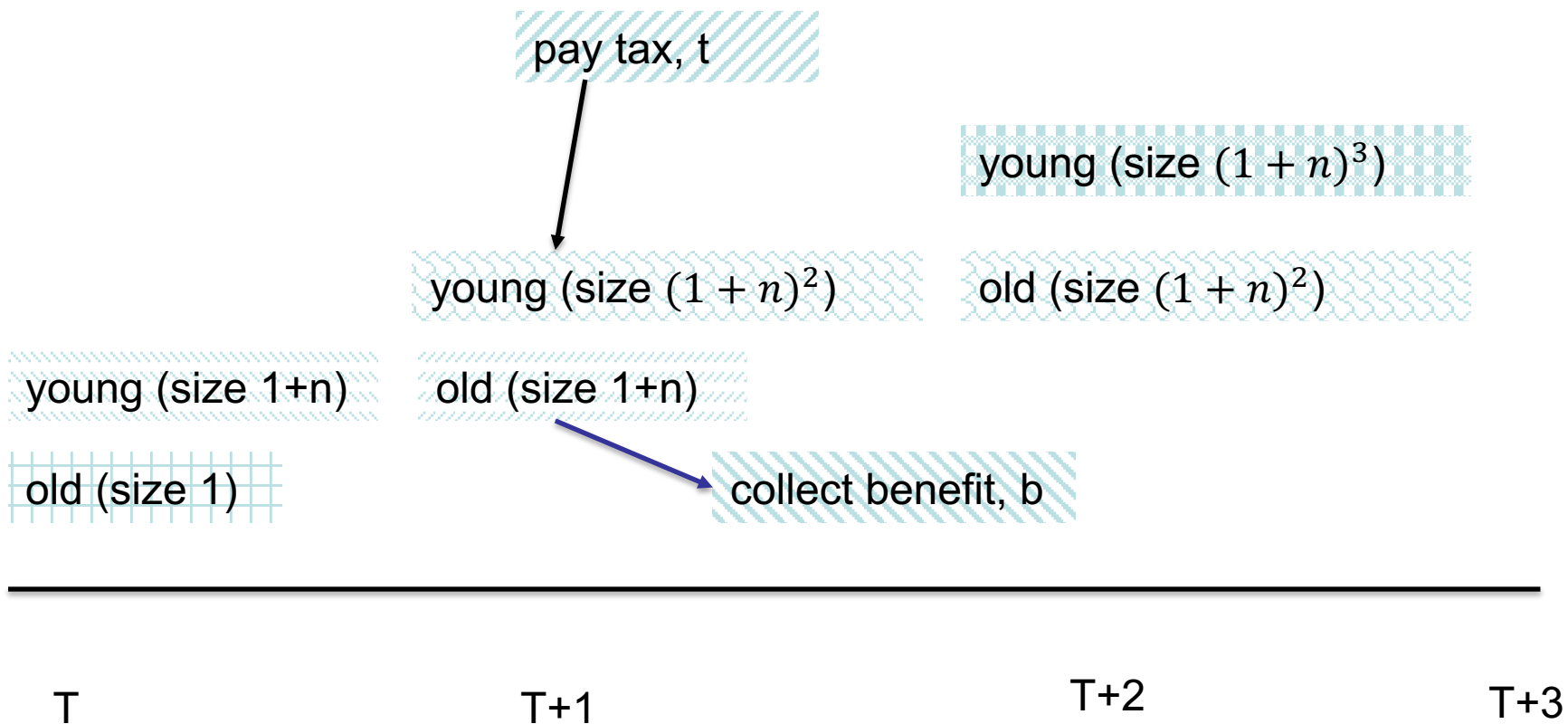
Population growth and economic growth



Aging population and pension

- Overlapping Generations Model
 - Each agent lives two periods: Young and Old
 - In any period, there are people of different generations alive
- Population Evolution:

$$N' = N(1 + n)$$



Aging population and pension

- Introduce Pay-As-You-Go Social Security:
 - Old: Social security payments b
 - Young: tax t

- Budget Equilibrium:

$$Nb = N't$$

- Total social security benefits must equal total taxes on the young

$$b / (1 + n) = t$$

- Social Security is a **forced savings program**.

Aging population and pension

- When Pay-As-You-Go Social Security was introduced, The First Generation of Old Consumer clearly benefited from the program:
 - they did not pay taxes when they were young
 - they now have an additional source of income
- For everyone else, the effect on welfare is less obvious

Aging population and pension

- Lifetime wealth of the consumer:

$$\begin{aligned} we &= y - t + \frac{y' - t'}{1 + r} \\ &= y - \frac{b}{1 + n} + \frac{y' + b}{1 + r} \\ &= y + \frac{y'}{1 + r} + \frac{b(n - r)}{(1 + r)(1 + n)} \end{aligned}$$

Aging population and pension

- Lifetime wealth of the consumer:

$$we = y + \frac{y'}{1+r} + \frac{b(n-r)}{(1+r)(1+n)}$$

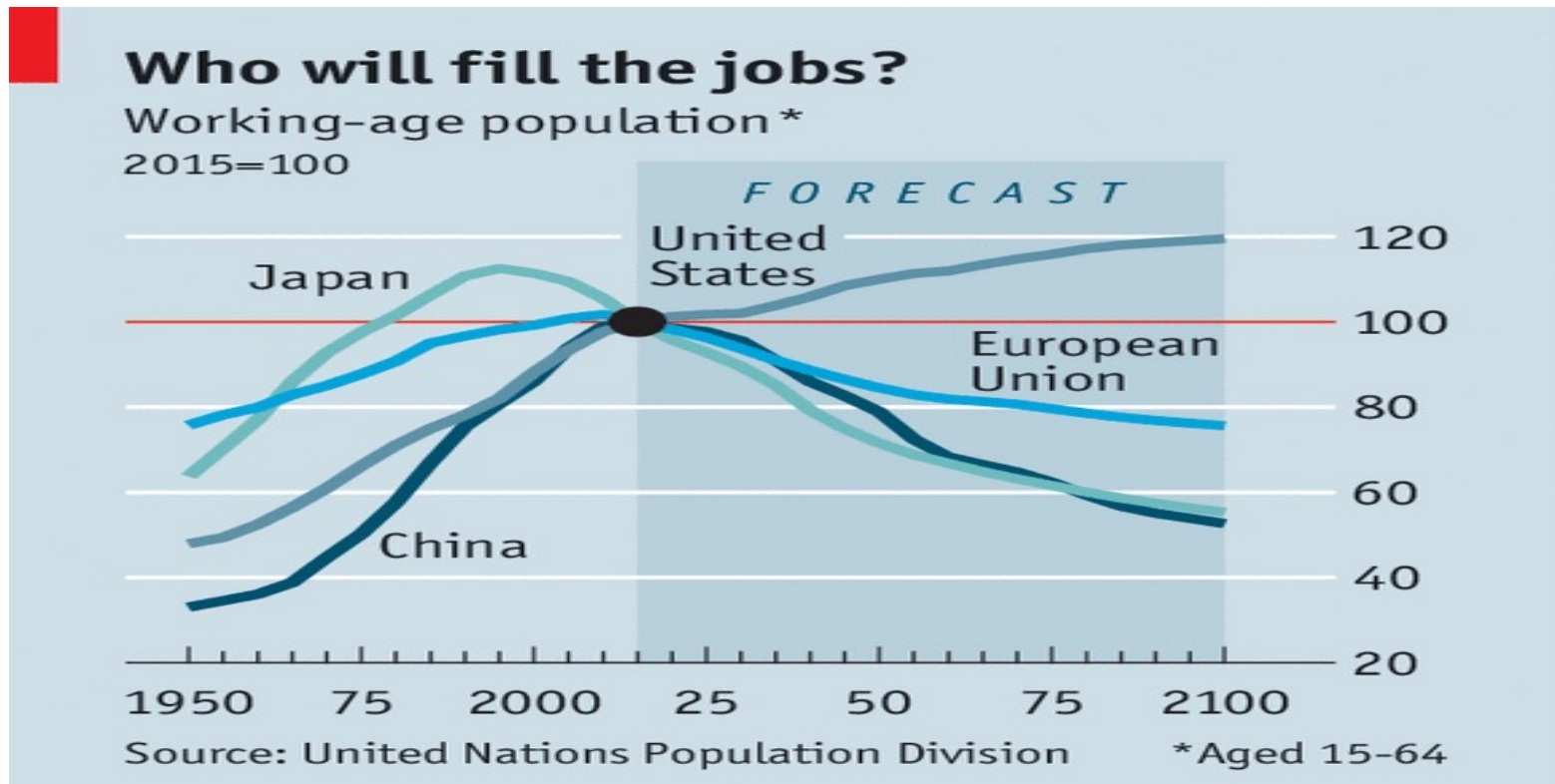
If $n > r$: Pay-As-You-Go Social Security improves welfare

If $n < r$: Pay-As-You-Go Social Security decreases welfare

Intuition:

- If people save by themselves for retirement, return on their savings equals $1 + r$
- If people save via a social security system return equals:
 $1 + n$

Vanishing workers



Economist.com

- <http://www.economist.com/news/finance-and-economics/21702477-can-debt-fuelled-model-growth-cope-ageing-populations-vanishing>

I. Demographic Measures

Birth rate (crude birth rate) : Birth per thousand of population.

Death rate (crude death rate) : deaths per thousand of population.

The rate of natural increase: Birth rate – Death rate

Example: 99 million at the start

5 million births

3 million deaths

101 million at the end

$$\text{Average population} = (99 + 101) \div 2 = 100$$

$$\text{Birth rate} = \frac{\text{Births}}{\text{Average population}} = \frac{5}{100} = 5\%$$

$$\text{Death rate} = \frac{\text{Deaths}}{\text{Average population}} = \frac{3}{100} = 3\%$$

$$\text{Natural growth rate} = \frac{5}{100} - \frac{3}{100} = \frac{2}{100} = 20\text{‰}$$

- Years required to double population:

$$\text{Doubling time} = \frac{70}{\text{Natural growth rate}}$$

- Example: growth rate 2%.

$$\text{Doubling time} = \frac{70}{2} = 35$$

- Infant death rate: deaths in the first year of life per thousand live births.

- Example: 10,000 born

1,000 deaths before birthdays

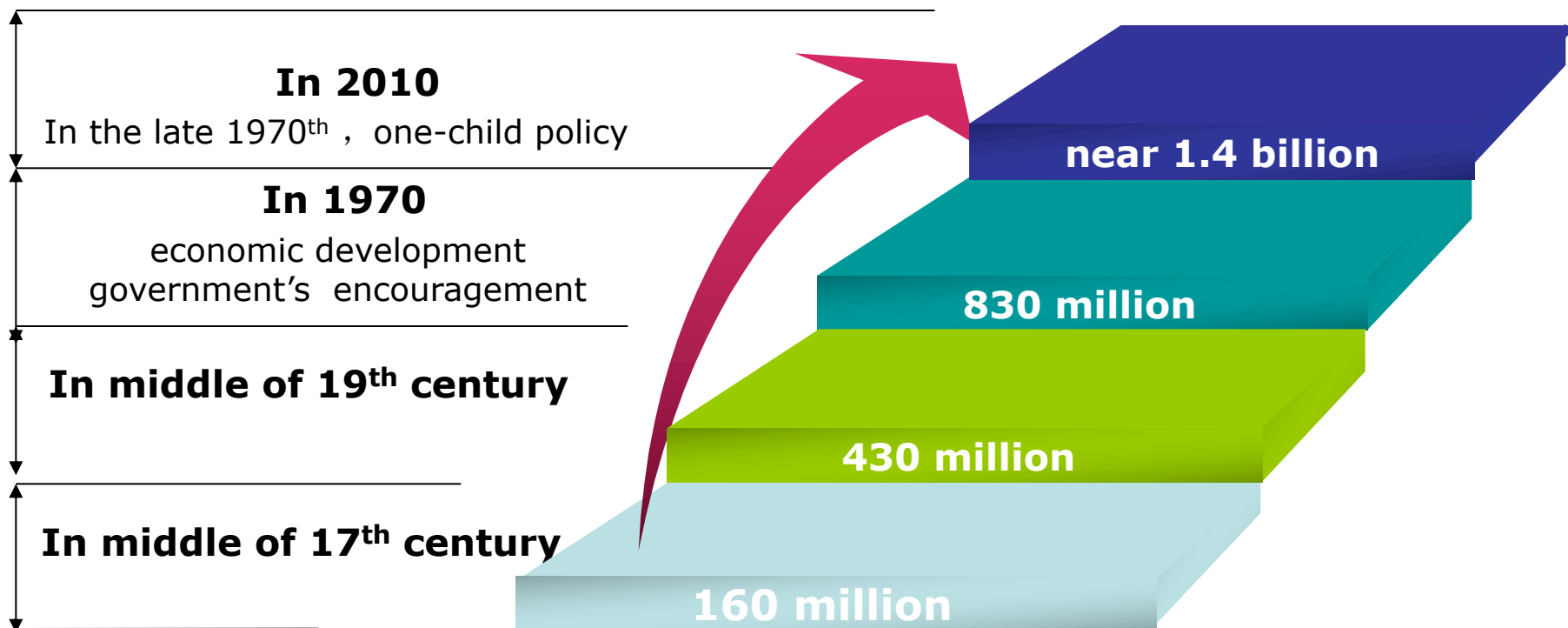
$$\text{Infant death rate} = \frac{1,000}{10,000} = \frac{10}{100} = 100\text{‰}$$

Rate of Natural Increase in Population

Country	2005-2010	2010-2015	Country/ region	2005-2010	2010-2015
Hungary	-0.4	-0.2	UK	0.2	0.6
Japan	-0.1	-0.1	USA	0.6	0.9
Sweden	0.2	0.6	China	0.7	1.2
Germany	-0.2	-0.2	Russia	-0.4	-0.1
India	1.4	1.3	Hong Kong	0.2	1.0
France	0.4	0.5	Mexico	1.4	1.1
Iraq	3.1	3.1	Yemen	4.7	3.0
Afghanistan		3.1	Malawi		3.2

Sources : United Nation: *Human Development Report* 2009 , 2011

China's population increased rapidly beginning from the Ching Dynasty



The Population in *China* 1–2008

(Unit:1000s)

year (AD)	Western Europe	China	Other	Total Asia (excluding)	World	/World (%)
1	24700	59600	36600	171200	230820	25.82
1000	25413	59000	41400	175400	267573	22.05
1300	58353	100000				
1400	41500	72000				
1500	57268	103000	55400	268400	438428	23.49
1600	73778	160000	65000	360000	556148	28.77
1700	81460	138000	71800	374800	603490	22.87
1820	133040	381000	89400	679400	1041834	36.57
1870	187504	358000	119792	730792	1271915	28.15
1913	260975	437140	184849	925689	1791091	24.41
1950	304941	546815	392827	1298642	2524324	21.66
1973	358825	881940	677613	2139553	3916489	22.52
2001	392101	1275392	1227630	3526612	6149006	20.74
2007		1318000	1560100	4010000	6625000	19.89
2008		1324655			6697254	19.78

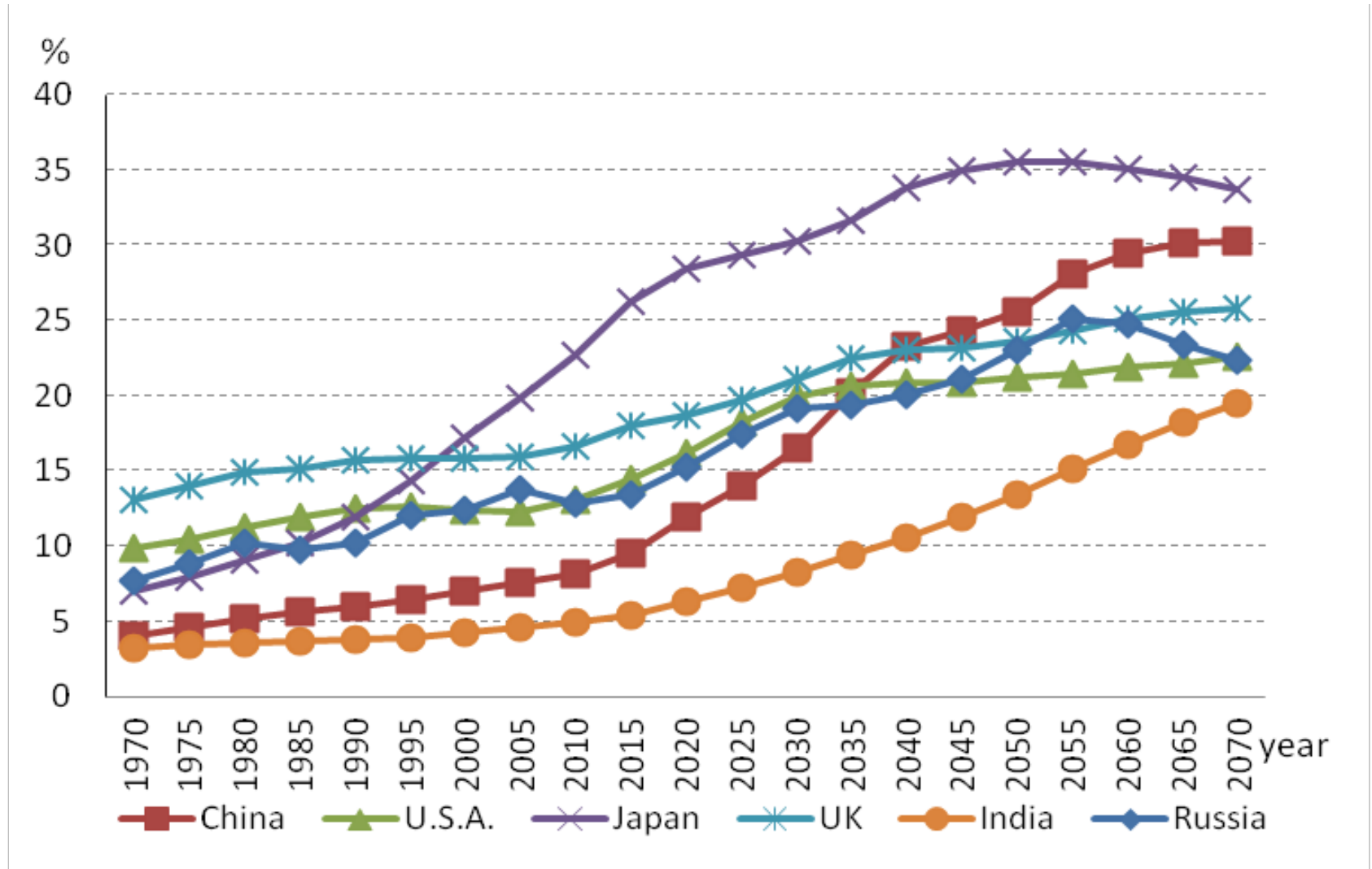
Sources: Maddison, Angus;
The World Economy: A Millennial Perspective Table 1–8a.

Proportion of the Population Age 65 or over (1950-2100)

	China	India	Japan	UK	USA	Russia
1950	4.512	3.137	4.947	10.832	8.265	6.186
1970	4.025	3.251	7.032	13.028	9.836	7.689
1990	5.942	3.786	11.948	15.703	12.489	10.200
2000	7.006	4.230	17.182	15.810	12.380	12.407
2010	8.189	4.922	22.687	16.588	13.059	12.799
2020	11.992	6.306	28.390	18.713	16.213	15.211
2030	16.470	8.271	30.282	21.113	19.907	19.137
2035	20.209	9.368	31.607	22.429	20.598	19.407
2040	23.300	10.545	33.803	23.030	20.912	20.117
2060	29.474	16.732	35.067	25.104	21.917	24.787
2065	30.223	18.212	34.488	25.546	22.118	23.391
2070	30.241	19.542	33.743	25.800	22.598	22.399
2075	30.185	20.848	33.017	26.320	23.200	22.374
2080	30.080	22.048	32.546	26.699	23.716	22.830
2085	29.670	23.109	32.180	27.003	24.254	23.273
2090	29.121	24.016	31.893	27.279	24.785	23.353
2095	28.617	24.779	31.700	27.523	25.276	23.093
2100	28.222	25.423	31.600	27.711	25.772	22.922

Data sources: World Population Prospects. The 2010 revision .United Nations Population Division

The Trend of Population above 65 for Selected Countries (1970-2070)

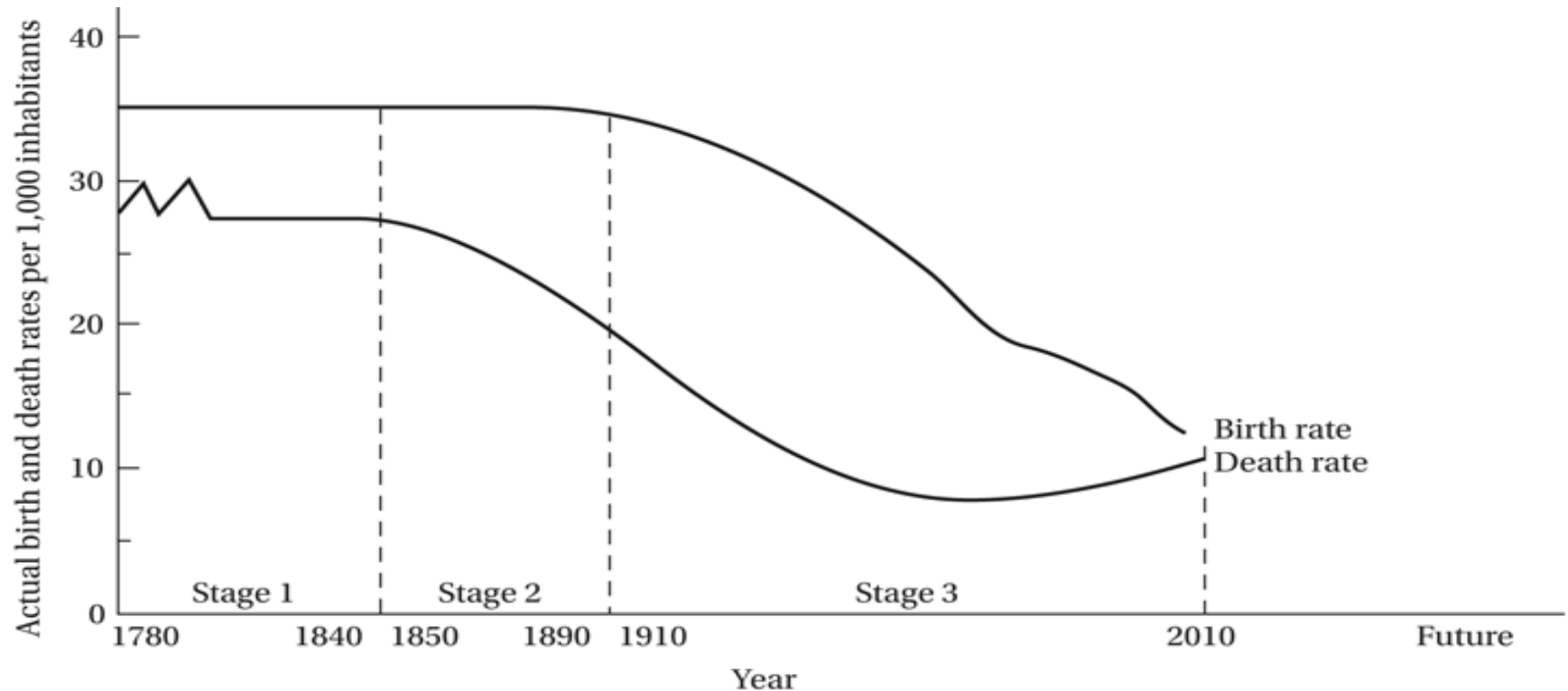


- Life expectancy:
 - Number of additional years that the average person of a given age will live age – specific death rates remain unchanged.
- Life expectancy at birth:
 - Number of years that an average new born person will live.
- Total Fertility rate:
 - Number of children the average woman will have in her lifetime. Age - specific fertility rate.

II. World population growth through history

Year	Estimated population (in millions)	Estimated Annual growth rate(%)
10,000 B.C	5	
1650	545	0.04
1850	1,171	0.53
1900	1,608	0.65
1950	2,576	0.91
1970	3,698	2.09
1980	4,448	1.76
1990	5,292	1.73
2000	6,260	1.70

- The demographic transition:
 - The process by which fertility rates eventually decline.



- Stage I: high birth rate, high death rate.
Stage II: beginning of the transition.
Stage III: decline in the birth rate.
Stage IV: population growth zero/negative.

III. Theory of Population Growth

A. Malthus' population trap: "Essay on the Principle of Population", 1798.

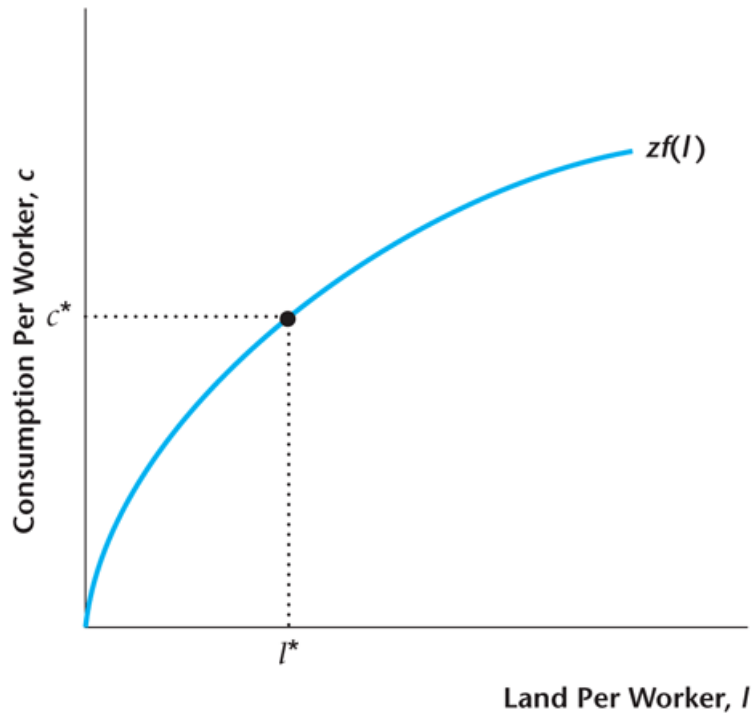
- a. The passion between the sexes would cause population to expand as long as food supplies permitted. Population would grow at a geometric rate.
- b. Land is fixed, marginal product of labor will decrease. Food would grow at an arithmetic rate.
- c. Low income and a stable population growth.
- d. Famines, wars, and epidemics reduce population.
 - "positive check"
 - A later marriage, "preventive check"
 - He was a minister, did not mention birth control, did mention "moral restraint".

- Higher population growth, the higher is per-capita consumption.

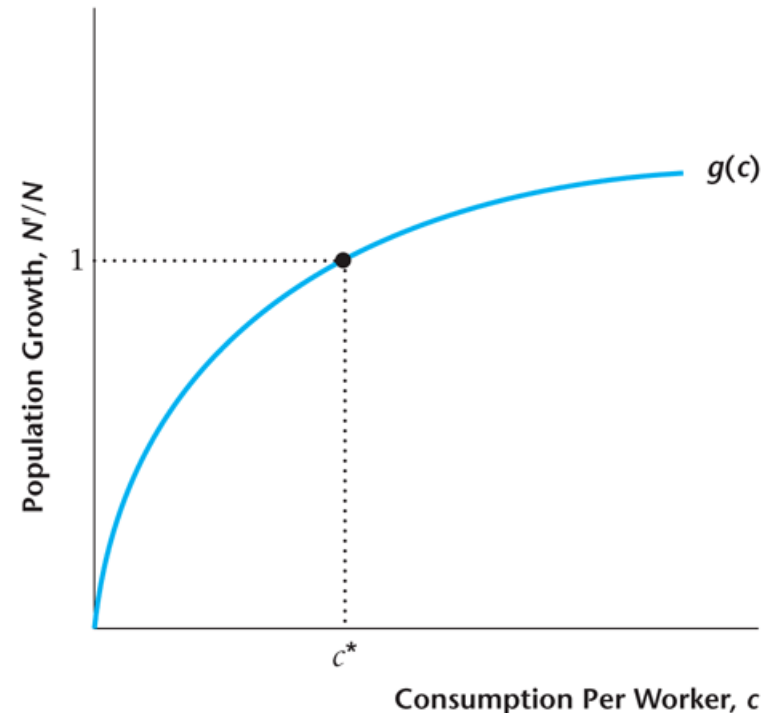
$$N' / N = g(C / N)$$

- Birth rate increases with consumption per capita
 - people will have more children by choice.
 - better nutrition increases fertility.
- Death rate decreases with consumption per capita
 - decreases infant mortality.
 - population healthier, increasing the average lifespan.

Determination of the Steady State in the Malthusian Model



(a)



(b)

- If z increases, this shifts up the per-worker production function.
- In the long run, the population increases to the point where per capita consumption returns to its initial level.
- There is no long-run change in living standards.

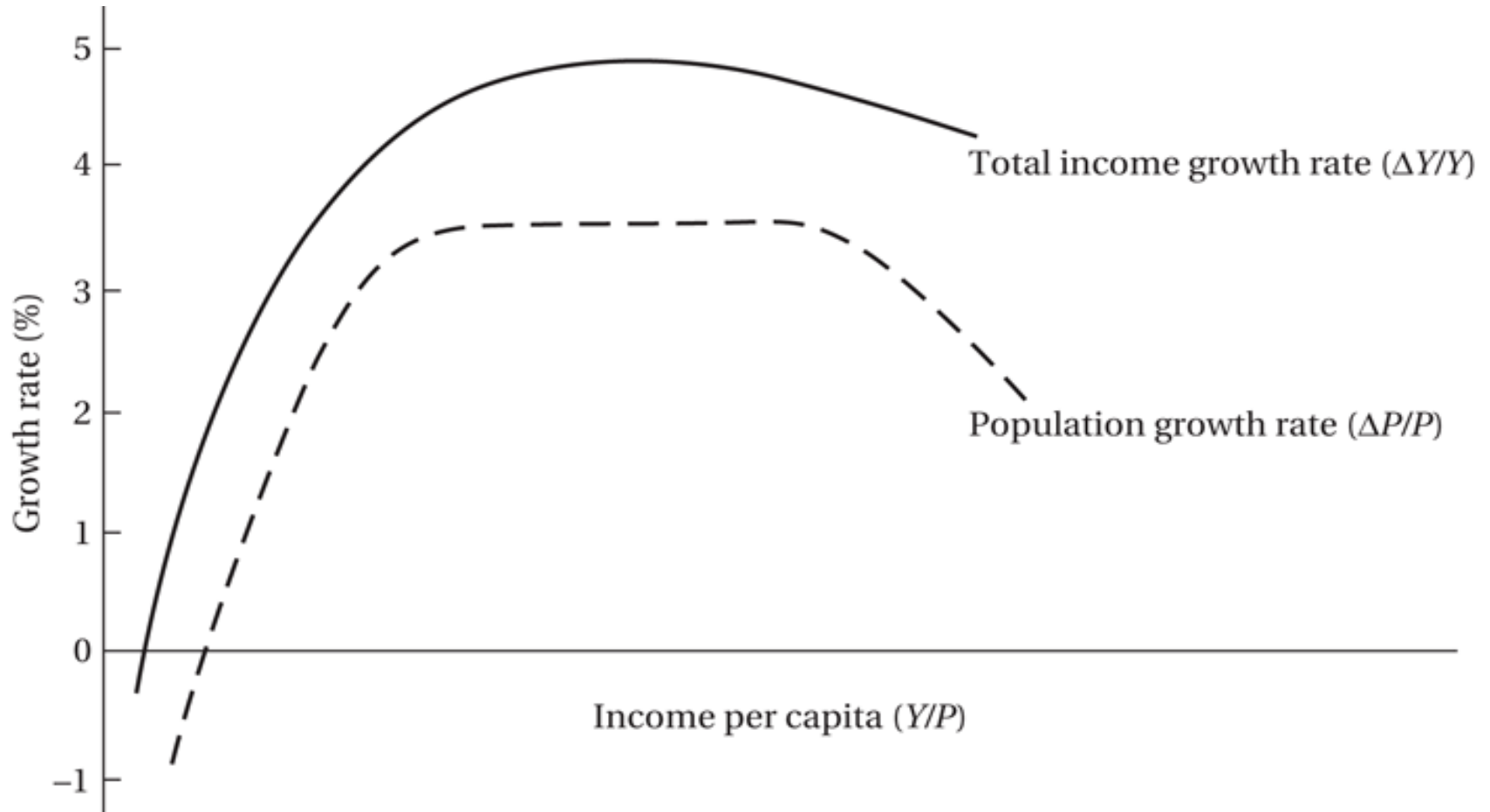
z increases

High yield,
more food,
Higher living standard,

more food,
higher population growth,
Less land per capita,
Lower yield per capita

- Criticisms of the Malthusian Model
 - Impact of technological progress
 - Currently no positive correlation between population growth and levels of per capita income in the data
- Microeconomics of family size; turns focus to individual rather than aggregate variables

How Technological and Social Progress Allows Nations to Avoid the Population Trap

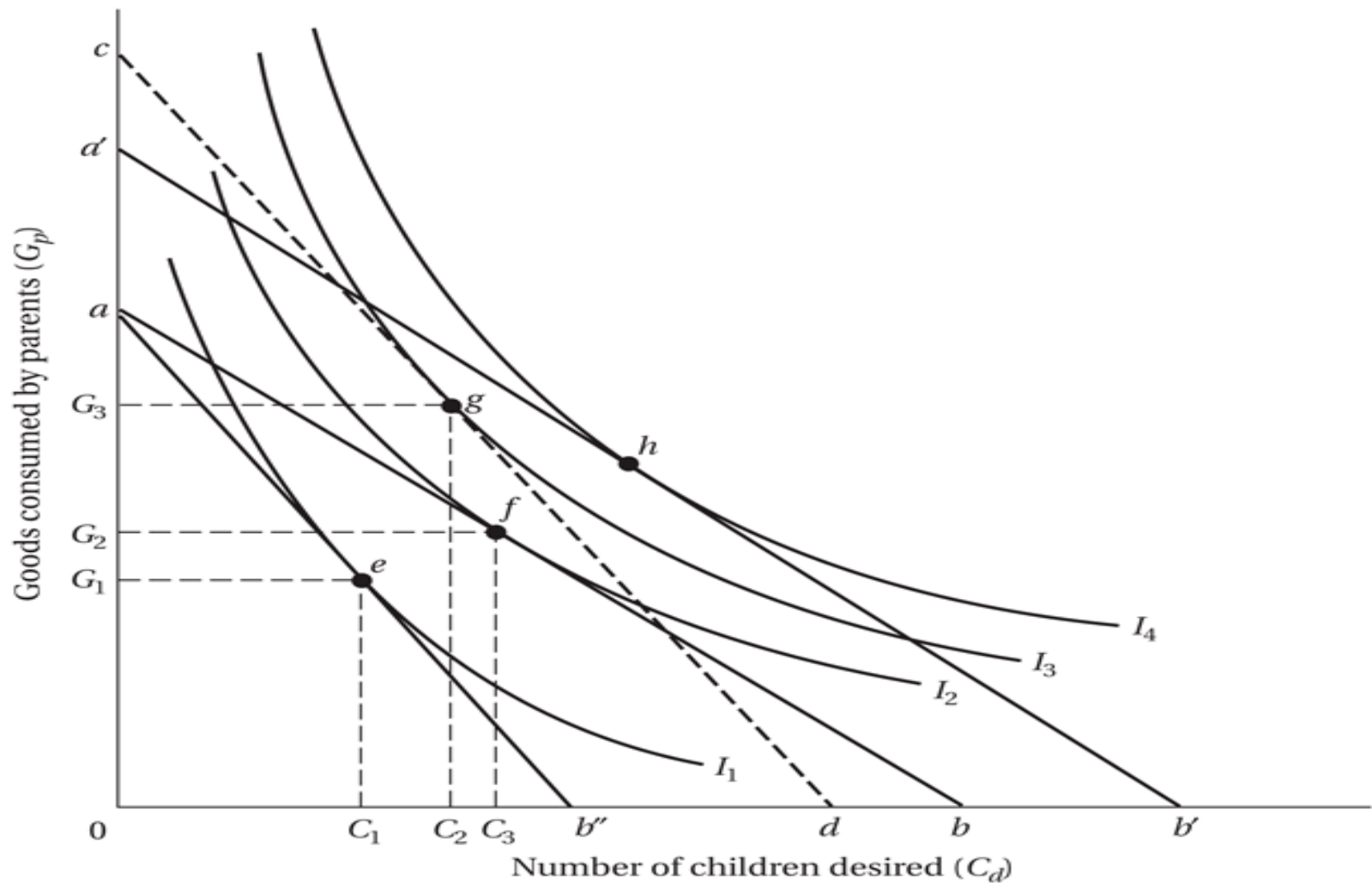


B. Becker's "New Household Economics"

- People are rational.
- Children are durable consumption goods.
- Couples maximize utility by choosing quantity of children.
- Utility depends on :
 - number of children,
 - quality of each child.
 - cost: time spent on rearing children.
 - some argued that parents' utility is equal to the worst child's utility.

- The Microeconomic Household Theory of Fertility
- The Demand for Children in Developing Countries
 - First two or three as “consumer goods”
 - Additional children as “investment goods”:
 - Work on family farm, microenterprise
 - Old age security motivation

Microeconomic Theory of Fertility: An Illustration



Demand for Children Equation

$$C_d = f(Y, P_c, P_x, t_x), x = 1, \dots, n$$

where

C_d : is the demand for surviving children

Y : is the level of household income

P_c : is the “net” price of children

P_x : is price of all other goods

t_x : is the tastes for goods relative to children

- The higher the household income, the greater the demand for children.
- The higher the net price of children, the lower the quantity demanded.
- The higher the prices of all other goods relative to children, the greater the quantity of children demanded.
- The greater the strength of tastes for goods relative to children, the fewer children demanded.

Lessons from Microeconomic Household Models

Fertility may be lowered with:

- Improved women's education, role, and status
- Female nonagricultural wage employment
- Rise in family income levels through shared growth
- Reduction in infant mortality, better health care
- Development of old-age and social security plans
- Expanded schooling opportunities, lowered real costs
- Lowered prices and better information on contraceptives
- Direct incentives such as subsidy benefits
- Policies that have the effect of reducing boy preference

The above list provides a framework for policy.

Some conflicting perspectives

- Population growth: “It’s Not a Real Problem” :
 - The real problem is not population growth but the following,
 - Underdevelopment
 - World resource depletion and environmental destruction
 - Population Distribution
 - Subordination of women
- “Overpopulation is a Deliberately Contrived False Issue”
- “Population Growth is a Desirable Phenomenon”

Some conflicting perspectives

- “Population Growth *Is* a Real Problem”
 - Extremist arguments
 - Theoretical arguments
 - Empirical arguments
 - Lower economic growth
 - Poverty
 - Adverse impact on education
 - Adverse impact on health
 - Food constraints
 - Impact on the environment
 - Frictions over international migration

Toward a Consensus

- Despite the conflicting opinions, there is some common ground on the following:
 - Population is not the primary cause of lower living levels, but may be one factor
 - Population growth is more a consequence than a cause of underdevelopment
 - It's not numbers but quality of life
 - Market failures: potential negative social externalities
 - Voluntary decreases in fertility is generally desirable for most developing countries with still-expanding populations

IV. Some policy approach

– What Developing Countries Can Do

- Persuasion through education
- Family planning programs
- Address incentives and disincentives for having children through the principal variables influencing the demand for children
- Coercion is not a good option
- Raise the socioeconomic status of women
- Increase employment opportunities for women (increases opportunity cost of having more children, as in microeconomic household theory)
- Help facilitate genuine and faster development of developing countries that still have high fertility rates

- What the Developed Countries Can Do Generally
 - Address resources use inequities
 - More open migration policies

- How Developed Countries Can Help Developing Countries with Their Population Programs
 - Research into technology of fertility control
 - Financial assistance for family planning programs

Case Study: China and India

- China and India are the world's two most populous nations, with 1.4 and 1.3 billion people respectively.
 - It is projected that their population will be 1.28 and 1.64 billion in 2065.
- Population policy in China
 - In 1980s, China initiated "one child policy".
 - By the mid-1990s, China's fertility rate reached 1.9 births per woman, and it fell further to below 1.6 by 2013.
 - It is estimated that in 2010, there were 106 males for every 100 females.
 - By 2050, China will have almost twice as many people above age 50 as below age 20.
- Population policy in India
 - Prime Minister Indira Gandhi tried to implement drastic population control in 1975-1977, with little success.
 - However, family planning did become more widely practiced due to: rising income, policy incentive to encourage smaller family.
 - A preference for boys over girls developed, particularly in the "Hindi Belt".